

1. (5 pts)

a) What is the dimension of a vector space?

The dimension of V is the number of vectors in any basis for V .

b) What is the rank of a matrix A ?

Rank of a matrix A is the number of pivots in an echelon form of A .

Also the dimension of the column space or row space of A .

2. (6 pts) Answer each question below and give a brief explanation.

a) Is it possible for a nonhomogeneous system of seven equations in six unknowns to have a unique solution for some set of right-hand side of constants?

Yes. This will happen if there are pivots in all six columns and the vector of constants is in the column space of A .

b) Is it possible for the system in part a) to have exactly 6 solutions for some set of right-hand side constants?

No. Any linear system has either no solution, a unique solution, or infinitely many solutions.

3. (4 pts) Let A be a 7×5 matrix.a) If A has a 2-dimensional null space, what is the dimension of the row space of A ?

Nullity of $A = 2$. Rank $A + \text{nullity of } A = 5$ since there are 5 columns. Therefore Rank $= 3 = \text{Dim of the row space}$

b) If A has a 2-dimensional null space, what is the dimension of the null space of A^T ?

Rank $A = 3 = \# \text{ of pivots in } A = \# \text{ of pivots in } A^T$
 so Rank $A^T = 3$ and $\# \text{ of columns in } A^T = 7$.
 Therefore dim of nullspace for A^T is $7 - 3 = \boxed{4}$